

## Scalability & Future Plan

|               |                                                             |
|---------------|-------------------------------------------------------------|
| Date          | 15 June 2026                                                |
| Team ID       | SWTID-2026-7169                                             |
| Project Name  | OptiCrop: Smart Agricultural Production Optimization Engine |
| Maximum Marks | 1 Mark                                                      |

### Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

### Current System Limitations:

| S.No | Limitation                                              | Impact                                                                | Priority to Address (High / Medium / Low) |
|------|---------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------|
| 1    | Runs only on Flask's single-threaded development server | Cannot handle high concurrent traffic in production                   | High                                      |
| 2    | Model trained on a static, fixed dataset                | Recommendations may not reflect newer regional/climate data over time | Medium                                    |
| 3    | No database — relies on flat CSV and pickle files       | Harder to track prediction history or scale data management           | Medium                                    |

### Scalability Plan:

| S.No | Scalability Aspect | Current State                                  | Proposed Upgrade / Solution                                                                |
|------|--------------------|------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1    | User Load          | Single local Flask dev server, low concurrency | Deploy with Gunicorn/Nginx or migrate to cloud hosting (AWS/Heroku) with load balancing    |
| 2    | Data Storage       | Static CSV dataset + pickle model file         | Migrate to a proper database (PostgreSQL/MongoDB) for prediction logs and larger datasets  |
| 3    | Performance        | Lightweight model, fast on small scale         | Introduce caching and asynchronous request handling for higher throughput                  |
| 4    | Security           | No authentication, minimal input validation    | Add input sanitization, rate limiting, and optional user authentication for production use |

### Future Roadmap:

| Phase | Planned Feature / Enhancement | Target Timeline | Expected Impact |
|-------|-------------------------------|-----------------|-----------------|
|-------|-------------------------------|-----------------|-----------------|

|         |                                                                                          |         |                                                                     |
|---------|------------------------------------------------------------------------------------------|---------|---------------------------------------------------------------------|
| Phase 2 | Cloud deployment (AWS/Heroku) with production WSGI server                                | Q3 2026 | Improved scalability and public accessibility                       |
| Phase 3 | Integration of additional ML models (Random Forest, KNN) with model comparison dashboard | Q4 2026 | Higher prediction accuracy and flexibility                          |
| Phase 4 | Mobile-friendly UI and multilingual support for regional farmers                         | Q1 2027 | Wider accessibility and adoption among non-English speaking farmers |